QUESTION ONE: Bike Riding

Sam and Trevor live in Christchurch. Because the city area where they live is flat, the two boys ride their bikes everywhere rather than spend money on bus tickets.

(a) Sam has to ride 2.4km to school and it normally takes him about 6 minutes. Calculate Sam's average speed in ms^{-1} riding to school.

(b) Sam learns in Science that while he is riding there are four main forces acting on him. These are: friction, thrust, gravity and reaction forces.

(i) On the diagram below draw in four labeled arrows to represent these four forces.



(ii) Describe a situation in which the two forces acting in the vertical direction would not be equal and opposite. Also comment on what would happen in this situation you have described.

(c) Trevor complains to Sam that he seems to have to do more work riding his bike than Sam does. Ignoring their fitness levels, suggest two reasons why Trevor may in fact be correct.

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(d) (i) Sam decides to calculate his maximum acceleration from a standing start. His speedometer tells him he had reached a speed of $20ms^{1}$ after 3.6s. Showing the equation you use and the working involved, prove that his acceleration was $5.6ms^{-2}$.

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(ii) The combined mass of Sam and his bike is 70kg. Assuming that Sam's maximum acceleration is $5.6\ ms^{-2}$, calculate the maximum force he uses to accelerate. Use the equation F = m x a.

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(iii) Trevor does the same calculation as Sam and finds that his maximum accelerating force is 340N. If Trevor applies this force steadily over a distance of 25m, calculate his power output if it took him 10s to travel the 25m.

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QUESTION TWO: Bungy Jumping

Sam decides to do the bungy jump while visiting Skipper's Canyon with his friend Trevor. The drop is around 100m from a bridge above the canyon. As he jumps, Sam is in free fall until the bungy cord begins to operate, and as it stretches, he slows down and stops just above the water surface before being bounced back up by the stretched bungy cord.

(a) Sam has a mass of 62kg. What is his weight?

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(b) Sam is in free-fall for 2.5 seconds before the bungy cord begins to stretch. If the acceleration due to gravity is $10ms^{-2}$, calculate his velocity at this point.

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(c) Would Sam's velocity that you calculated in (b) be Sam's maximum velocity during his fall? Explain your answer.

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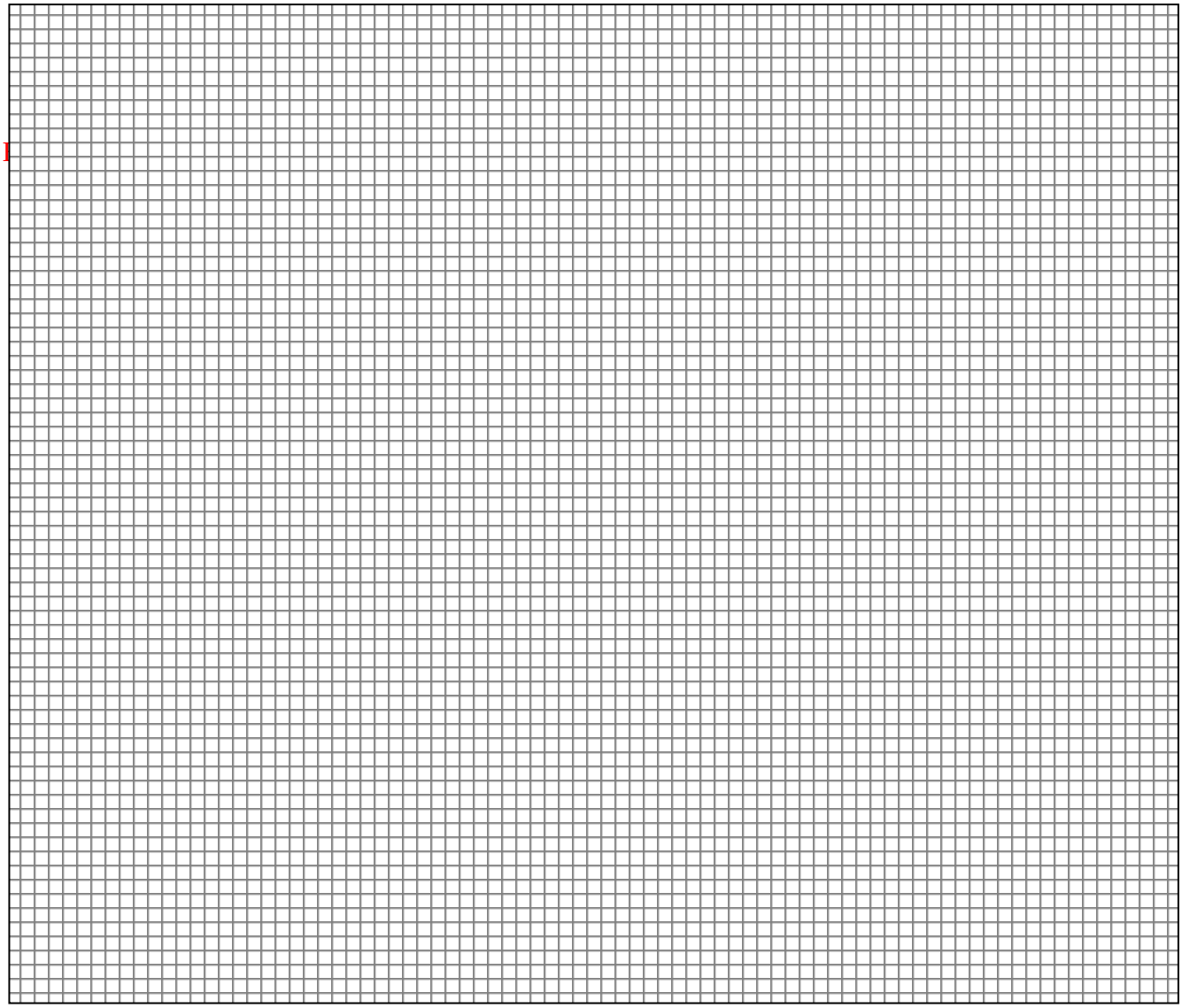
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(d) In the space below sketch a speed-time graph of Sam's motion from the bridge to the water. Label your axes but do not worry about a scale or units. Clearly identify the position of 2.5 seconds. (A)(M)(E)

(e) While he is watching, Trevor decides that when he gets home he will figure out exactly how high the bridge is. He drops a stone over the side of the bridge and records on his watch how long it took to fall. He recorded 4.6s. When he gets home, Trevor draws a graph of the motion of the stone and analyzes it for the distance fallen.

(i) Using the graph paper provided below, construct your own graph of the stone's motion. Use the fact that the speed of the stone increases by $10ms^{-1}$ every second and put speed on the vertical axis and time on the horizontal.



(ii) By clearly marking your graph show how Trevor would have calculated the distance the stone fell.

(iii) Calculate this distance from your graph.

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(f) Using data from earlier in Question Two, state the size of the force of gravity acting on Sam?

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(g) Using the equation W = F x d, calculate the work done by gravity on Sam in a 100m fall.

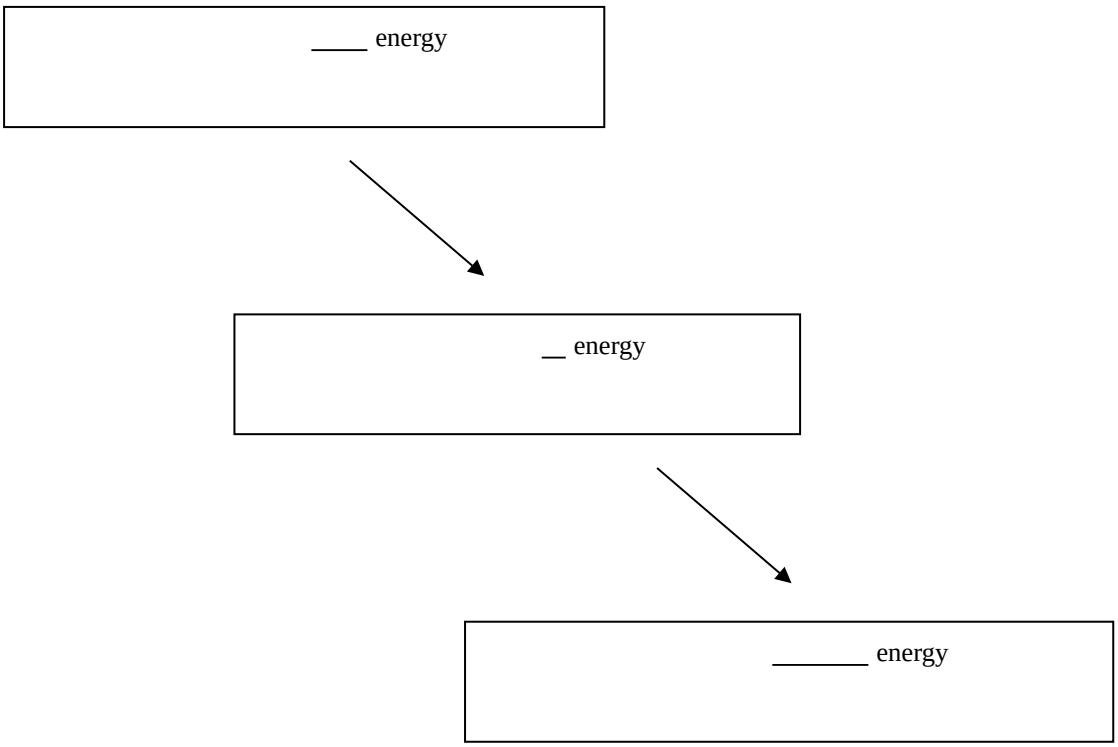
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(h) As Sam falls, list the three types of energy involved in their correct order.

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QUESTION THREE

(a) Mary is a keen follower of the national cricket team. She likes the black uniform they wear but from her understanding of radiant heat she feels that the colour black is not a good choice. She discusses it in class and realizes that the black colour of the uniform is both an advantage and also a disadvantage for the wearers. Describe how the black colour can be both an:

advantage _____

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_____(M)

and a disadvantage _____

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_____(A)

(b) Heat can be transferred by conduction, convection and radiation. For each of the following state which type of heat transfer is operating.

(i) The transfer of heat through outerspace _____

(ii) The transfer of heat through the bottom of a frying pan _____

(iii) The transfer of heat through the water in a hot water cylinder _____

QUESTION FOUR

A light bulb is rated at 100W and 240V.

(a) Calculate the resistance of the filament in the bulb and show that its value is 576Ω

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(b) In order to check this calculated value, Lapi constructed a circuit using a 12V battery, the light bulb, a voltmeter and an ammeter, and recorded the readings on both meters.

(i) Using the electrical symbols supplied at the front of this paper, draw a circuit diagram of Lapi's experiment.

(ii) Lapi's results were as follows:

Voltmeter = 11.8V

Ammeter = 22mA

Because the current was so small, Lapi had to use a milliammeter for the reading. What is the current reading in amps? _____

Using Ohm's Law; V = I x R, calculate the value of the resistance of the light bulb from Lapi's results.

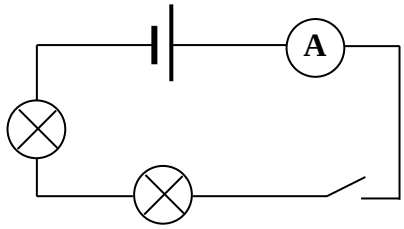
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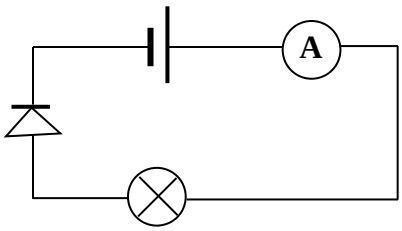
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(c) Lapi was expected to construct the circuits represented by the following circuit diagrams.

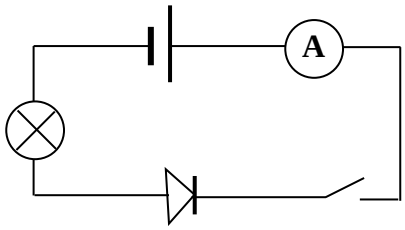
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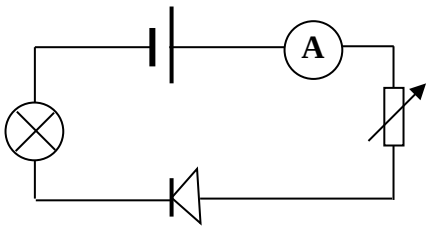
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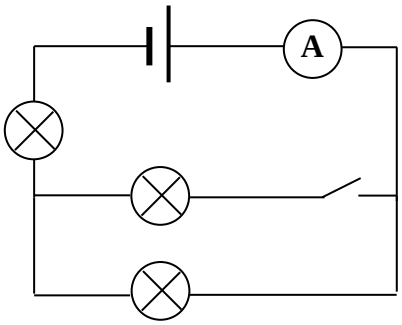
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(i) In which circuit is there a parallel connection? _____

(ii) In which circuits is there opportunity to vary the current flowing through the ammeter. (Ignore zero readings) _____

(iii) In which circuits is the ammeter reading zero? _____

(iv) In which circuit will there be no current even with the switch closed? _____